Bus Protocols

OR

Low Level Wired Serial Communication Protocols

AGENDA

- What is Embedded System ?
- Need for Serial Communication Protocols in ES.
- Intra Circuit vs Inter Circuit Communication ?
- What is Network ?
- What is a Protocol ?
- Low Level /BUS Protocol vs High Level Protocol.
- Bit-Banging Method vs Peripheral Method of Bus Protocol Implemention ?
- Embedded Networking.
- Introduction to I2C protocol.

- What is Network ?

  Network refers to the interconnection of two or more devices wired/wireless, directly/indirectly to accomplish reliable communication using one or more protocols functioning between them.

- What is protocol ?

  Refers to a set of rules, regulations , checkpoints & sometimes standards to be followed for reliable communication.

- Low Level protocols involves rules,regulations,&checkpoints(logical aspect) & standards(physical aspect) as well.

Examples:

Wired protocols (Bus Protocols):

UART's, I2C, SPI, 1-Wire, CAN, LIN, Modbus, USB, etc

Wireless protocols:

(RC5,RC6,Sony,etc)-Remote Control ,BlueTooth,RF,WIFI,NFC,etc

- High level protocols :

  Involves ONLY rules , regulation & checkpoints (logical aspect).

  BUT Standards (Physical Aspects) ARE NOT A Concern.

  Hence Depend On LOW LEVEL Protocols to Convey Information Across Physical Wired/Wireless Medium.

- What is protocol suite ?

  Refers to set/collection/bunch/stack/layers of protocols (BUT NOT just ONE protocol) being used for communication between connected devices.

  Examples:

  TCP/IP,AUTOSAR/CANOpen,MilCAN,CANAerospace,RJ1939,CoAP, etc

- Embedded Networking

  Communication ,between two or more embedded devices(inter-circuit) ,or between 2 or more CPUs(intra circuit),or between CPU(s) & certain peripheral components on the same embedded board(intra circuit) can be referred to as embedded networking.

  Wired & wireless protocols may be involved.

I2C Bus Protocol Introduction

1. The acronym I2C stands for “Inter Integrated Circuit” - is a synchronous serial communication protocol.
2. This protocol was suggested by Philips in the early '80s to allow easy communication between components which reside on the same circuit board.Hence MOSTLY INTRA-CIRCUIT COMMUMICATION.
3. It supports a 2 wired bidirectional bus, with the bus lines called as scl & sda.

   scl – is serial/synchronous/shift clock line

   sda –is the serial data line for transmission or reception.
4. It supports ONLY half-duplex serial communication.
5. The bus length can be between 9 to 12 feet.

6. It is a “Defacto (Well established & organized) Standard” .
7. It support Bus speeds of the order

- 100 kbits/s or KHz – standard mode
- 400 kbits/s or KHz – fast mode (fm)
- 1 Mbits/s or MHz -- fast mode plus (fm+)
- 3.4 Mbits/s or MHz – High speed (HS) mode
- 5 Mbits/s or MHz – Ultra High-speed/Fast mode(UfM)

8. Supports Multi-Master-Slaves communication.
   Master is always any CPU(uC usually)
   Slaves are mostly peripheral devices
   BUT sometimes can also be other CPU(s).

9. Supports ONLY MSB first serial data transfer order between master & slaves.

10. Supports Handshaking in the form of acknowledgement signal/state for every byte communicated between master & slave.

11. Supports slave access/selection/identification using addressing method .There are 2 ranges:

    7-bit slave addressing scheme &

    10-bit slave addressing scheme.

12. Supports Broadcast & Unicast Communication ONLY